

Probability of an Event

- We consider only those experiments for which the sample space contains a finite no. of elements.
- The likelihood of the occurrence of an event resulting from such a statistical experiment is evaluated by a means of a set of real nos., called cuts. & probabilities, that ranges from 0 to 1.
- To every pt. in the sample space, we assign a probability such that sum of all probabilities is 1.
- To find the probability of an event A , we sum all the probabilities, assigned to the sample points in A . This sum is defined as the probability of A and is denoted by $P(A)$.

$$0 \leq P(A) \leq 1, \quad P(\emptyset) = 0 \quad \text{and} \quad P(S) = 1$$

- If A_1, A_2, A_3 is a sequence of mutually exclusive event, then

$$P(A_1 \cup A_2 \cup A_3 \cup A_4 \dots) = P(A_1) + P(A_2) + \dots$$

- If the sample space for an experiment contains N elements, all of which are equally likely to occur, we assign a probab. of $1/N$, to each of the N points.

- If an experiment can result in any one of the N different equally likely outcomes, and if exactly n of these outcomes correspond to event A , then,

$$P(A) = \frac{n}{N}$$

- If the outcomes of an experiment are not equally likely to occur, the probabilities must be assigned on the basis of prior knowledge or experimental evidence.
- Let's say, if a coin is not balanced, then we can estimate the probabilities of heads & tails by tossing that coin a large no. of times.
- As per the relative frequency defⁿ of probab., the true probabilities would be the frac^{ns} of heads & tails that occur in the long run.
- The indifference approach allows you to determine the probab. of die (that you think is balanced) by assigning the probability of $1/6$ to each side of the die.
- The use of intuition, personal beliefs, and other indirect info. that might be used to arrive at the probabilities is referred to as - the subjective defⁿ of probability.

- In most of the applications of probability, the relative frequency interpretation of probability is generally used. Its foundation is the statistical experiment rather than subjectivity, and it is best viewed as the limiting relative frequency.

- When opinions & prior info. differ from individual to individual, the subjective probability becomes the relevant resource.

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